Circular motion

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Relate inward net force to a change in velocity direction.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 9.*
Equal fall times; faster ball travels 5 times farther horizontally.

02. Worked example: explain the model
Velocity tangent at each point; net force toward center, perpendicular to instantaneous velocity.

03. Read and interpret the model
Velocity tangent; acceleration inward. Gravity, tension, friction, etc depending context.

04. Apply the physics
6 N; doubles to 12 N.

05. Explain your reasoning
Initially along tangent, not radially outward; subsequent forces may change path.

06. Review: connect an earlier idea *Review: Unit 1, Day 20.*
$a=10\ m/s^2$ right; $v_f=70$ m/s right. Constant net force and mass throughout.

07. Exit check
No; direction changes.

08. STRETCH: test the model
Force becomes four times as large because speed is squared; 6 N becomes 24 N.

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01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 9.*
Speed 15 m/s. Need an initial position plus time and motion history/conditions; velocity alone is insufficient.

02. Worked example: explain the model
Velocity tangent at each point; net force toward center, perpendicular to instantaneous velocity.

03. Read and interpret the model
Velocity tangent; acceleration inward. Gravity, tension, friction, etc depending context.

04. Apply the physics
a 8 m/s², F 4 N. Doubling v quadruples: a 32, F 16 N.

05. Explain your reasoning
Initially along tangent, not radially outward; subsequent forces may change path.

06. Review: connect an earlier idea *Review: Unit 1, Day 20.*
x interval [-88, -72] km; y interval [90,110] km. Simplified geometry, gravity, Earth motion or missing observations limit the model. Bounds are not probabilities.

07. Exit check
Both halve; inverse proportional to radius.

08. STRETCH: test the model
$v_{max}=\sqrt{(Fr/m)}=4$ m/s; at 2 m, $\sqrt{32}\approx5.66$ m/s.